$$\int e^{at}\,dt = \frac{1}{a}e^{at} + k$$

No. 29 JANUARI 2023

Date: SELASA

1. Arus $i(t) = \frac{dq}{dt} = e^{-2t}$

a) Tentukan $q(t)$ pada $t=0$, $q=0$

$i(t) = \frac{dq}{dt}$

$\int i(t)\,dt = \int \frac{dq}{dt}\,dt$

$\int i(t)\,dt = q(t)$

$q(t) = \int e^{-2t}\,dt = \left(\frac{-e^{-2t}}{2} + k\right)$

$q(t) = -\frac{1}{2}e^{-2t} + \frac{1}{2}$

b) Tentukan $q(t)$ pada $t=0$, $q=5$

$q(t) = \int i(t)\,dt$

$q(t) = \int e^{-2t}\,dt = \left(\frac{-e^{-2t}}{2} + c\right)$

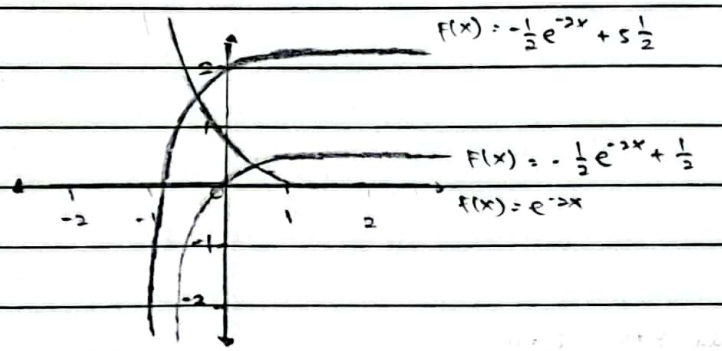
saat $q = 5 = c$

maka, $q(t) = \left(\frac{-e^{-2t}}{2} + 5\right)$

maka,

$q(t) = -\frac{1}{2}e^{-2t} + 5\frac{1}{2}$

c) Grafik :




Transien : steady : stabil

2. $\frac{dq}{dt} + 2q(t) = 0$

a) Tentukan $q(t)$ pada $t=0$, $q=0$

$\frac{dq}{dt} = -2q(t)$

$dq = -2q(t)\,dt$

$\int dq = \int -2q(t)\,dt$

$q + c_1 = -2t + c_2 = q(t)$

maka, $q(t) = 2t + c + q$ $\quad q = 0$

$q(t) = 2t + c$, $c$

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b). Tentukan q(t) pada t=0 dan q=5

$\frac{dq}{dt} = -2q(t)$ maka, $q(t) = 2t + C + q$ $\quad q = 5$

$q(t) = 2t + 5 + C$ //

$dq = -2q(t) \cdot dt$

$\int dq = \int -2q(t) \cdot dt$

$q + c_1 = -2t + c_2 + q(t)$

3. $\frac{dq}{dt} + 2q(t) = 10$

$\frac{dq}{dt} = 10 - 2q(t)$

$\int \frac{dq}{10 - 2q(t)} = \int dt$

$-\frac{1}{2}\ln(10 - 2q(t)) = t + c_1$

$\ln(10 - 2q(t)) = -2t + c_1$

$10 - 2q(t) = e^{-2t} \cdot (e^{c_1}) A$

$10 - 2q(t) = Ae^{-2t}$

$-2q(t) = -10 + Ae^{-2t}$

$q(t) = 5 - \frac{A}{2}e^{-2t}$ C //

4. $2\frac{di}{dt} + 10i = 7$

Tentukan arus i(t) pada t=0, i=0

$2\frac{di}{dt} + 10i = 7$ maka, $i = \frac{7}{10} + ke^{-5t}$

$2\frac{di}{dt} = 7 - 10i$

$\frac{di}{7 - 10i} = \frac{1}{2}dt$

$\int \frac{di}{7 - 10i} = \int \frac{1}{2} \cdot dt$

$-\frac{1}{10}\ln(7 - 10i) = \frac{1}{2}t + C$

$\ln(7 - 10i) = -5t + C$

$7 - 10i = e^{-5t} \cdot (e^{c}) k$

$7 - 10i = ke^{-5t}$

$-10i = -7 + ke^{-5t}$

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5. $2\frac{di}{dt} + 10i = 7e^{-t}$

Tentukan arus $i(t)$ pada $t=0$ dan $i=0$

$2\frac{di}{dt} + 10i = 7e^{-t}$

$\frac{di}{dt} = 7e^{-t} - 10i$

$\frac{di}{7e^{-t} - 10i} = \frac{1}{2}\,dt$

$\int \frac{di}{7e^{-t} - 10i} = \int \frac{1}{2}\,dt$

$-\frac{1}{10}\ln(7e^{-t} - 10i) = \frac{1}{2}t + c$

$\ln(7e^{-t} - 10i) = -5t + c$

$7e^{-t} - 10i = e^{-5t} \cdot e^{c}\ k$

$-10i = k e^{-5t} - 7e^{-t}$

$i = \frac{k e^{-5t} - 7e^{-t}}{-10}$